322674(22)

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BE (6th Semester)

Examination, April-May, 2016

(New Scheme)

Inter-Networking with TCP / IP

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of **all** questions is compulsory. Answer any **two** from parts (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) What is Internetworking ? Why is it required ? 2

(*b*) Explain the differences between OSI model and the TCP / IP Protocol suite. 7

(*c*) Define Gateway. Explain its role in computer networking. 7

(*d*) Explain the differences between the following : 7

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(*i*) Switch and Router

(*ii*) Repeater and Hub

2. (*a*) What is the purpose of Internet Protocol ? 2

(*b*) Draw and explain various fields of an IP datagram. 7

(*c*) Answer the following : 7

(*i*) What is the class of the following IP address ?

10101111 11000000 11110000 00011101

(*ii*) What is the net id and host id of the following IP address ?

190 . 57 . 82 . 6

(*iii*) Change the following IP address from dotted decimal notation to binary nottion :

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130 . 34 . 7 . 15

(*iv*) What is the subnet address if the IP address is 182.44.82.16 and the mark is 255 . 255 . 255 . 192 ?

(*d*) Explain ARP and RARP for dynamic mapping of logical address with physical address. 7

3. (*a*) In multicasting, there is :

(*i*) one source and one destination

(*ii*) one source and a group of destinations

(*iii*) one source and all hosts on the network

(*iv*) None of the above 2

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(*b*) Explain different types of packets used by OSPF routing protocol. 7

(c) Briefly explain MBONE. 7

(d) Explain Distance Vector Routing with example. 7

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4. (a) Regulating the amount of data a source can send before receiving an acknowledgement from destination is :

(i) Error Control

(ii) Flow Control

(iii) Conjestion Control

(iv) None of the above 2

(b) Explain differences between TCP and UDP headers. 7

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(c) Explain sliding window protocol used by TCP for flow control. 7

(d) Write short notes on the following : 7

(i) Leaky Bucket Algorithm

(ii) Token Bucket Algorithm

5. (a) Define Path Vector Identifier. 2

(b) Explain the concept of IP subnet and explain how binding of IP address is performed in an ATM network. 7

(c) Explain ATM reference model with suitable diagram. 7

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(d) Write short notes on the following : 7

(i) Packet-type and multiplexing

(ii) Connection Management

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